

$$(1) L \rightarrow EL'$$

$$L' \rightarrow EL' \mid \epsilon$$

(2)

	FIRST	FOLLOW
E	num, id, (	num, id, (, \$,)
A	num, id	num, id, (, \$,)
B	(	num, id, (, \$,)
L	num, id, (	)
L'	num, id, (, ϵ	)

(3) 对于有多个候选式的产生式

$$E \rightarrow A \mid B \quad \text{FIRST}(A) \cap \text{FIRST}(B) = \{\text{num}, \text{id}\} \cap \{(\} = \emptyset$$

$$A \rightarrow \text{num} \mid \text{id} \quad \text{FIRST}(\text{num}) \cap \text{FIRST}(\text{id}) = \emptyset$$

$$L' \rightarrow EL' \mid \epsilon \quad \text{FIRST}(EL') \cap \text{FOLLOW}(L') = \{\text{num}, \text{id}, (\} \cap \{) \} = \emptyset$$

所以(1)中得到的文法是LL(1)文法

(4)

	num	id	(	)	\$
E	$E \rightarrow A$	$E \rightarrow A$	$E \rightarrow B$		
A	$A \rightarrow \text{num}$	$A \rightarrow \text{id}$			
B			$B \rightarrow (L)$		
L	$L \rightarrow EL'$	$L \rightarrow EL'$	$L \rightarrow EL'$		
L'	$L' \rightarrow EL'$	$L' \rightarrow EL'$	$L' \rightarrow EL'$	$L' \rightarrow \epsilon$	

二、拓广文法

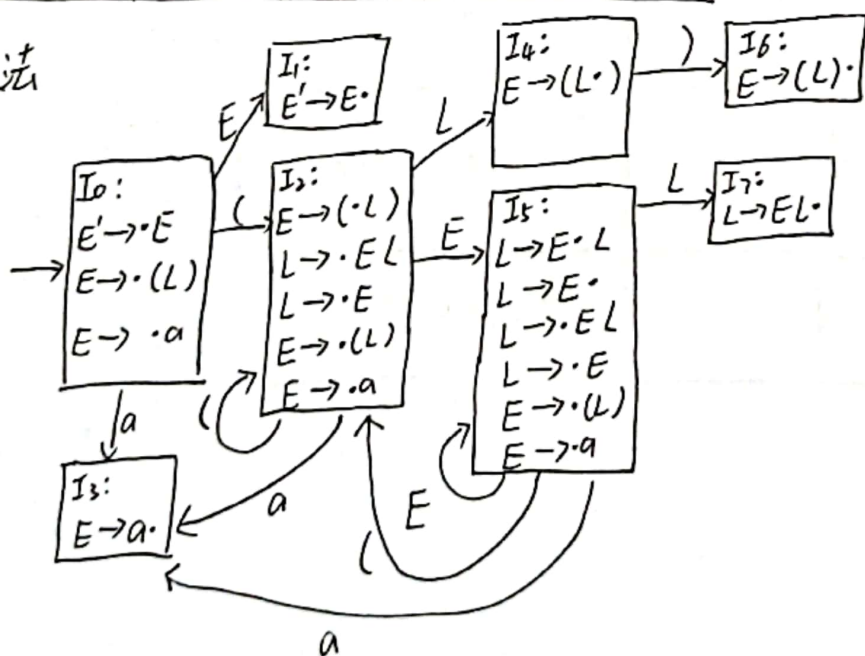
$$E' \rightarrow \bar{E}$$

$$E \rightarrow (L)$$

$$E \rightarrow a$$

$$L \rightarrow EL$$

$$L \rightarrow E$$



(2)

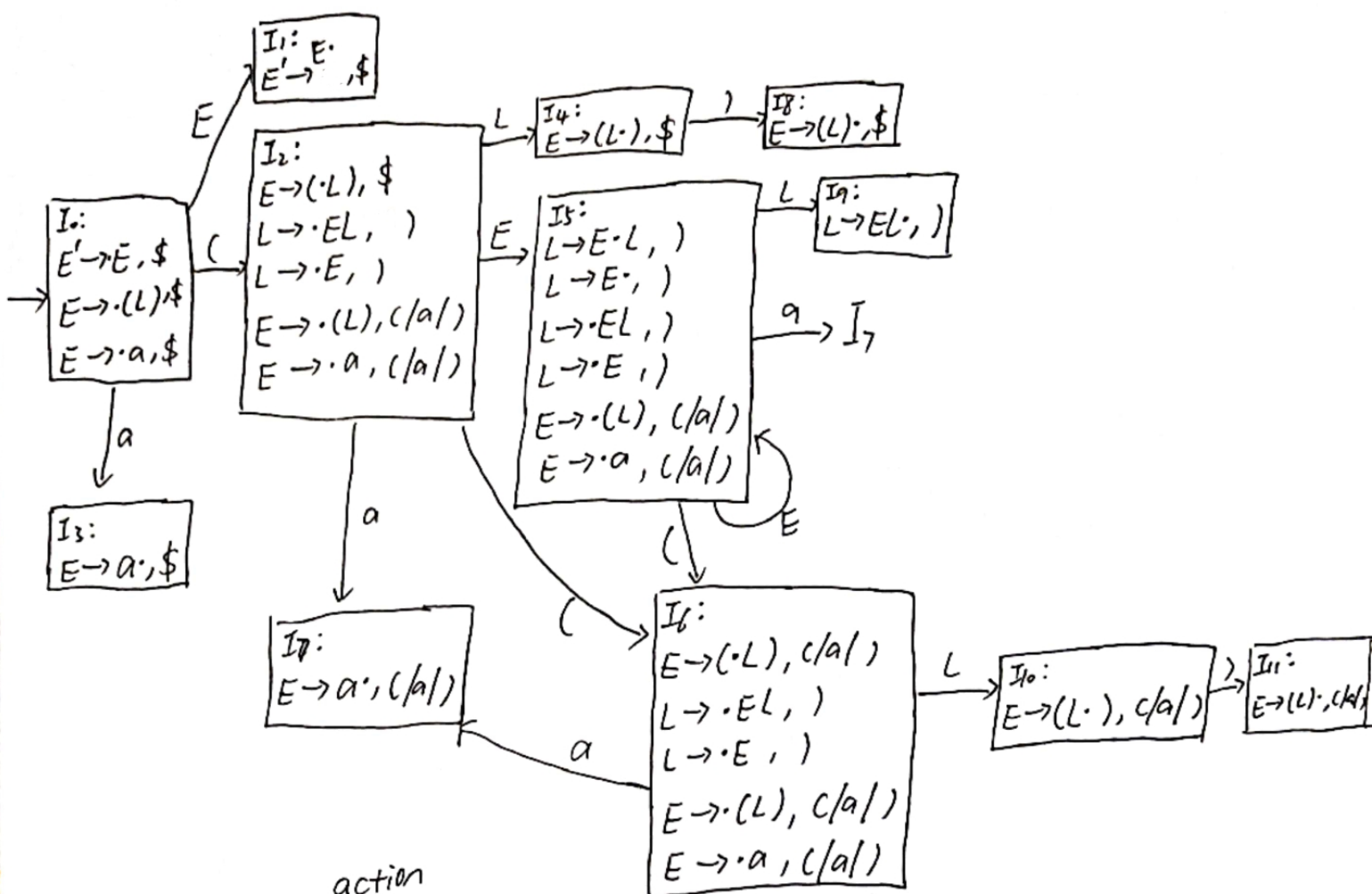
action				goto	
	(	)	a	\$	E L
0	S2		S3		1
1				ACC	
2	S2		S3		5 4
3	R(E→a)	R(E→a)	R(E→a)	R(E→a)	
4		S6			
5	S2	R(L→E)	S3		5 7
6	R(E→(L))	R(E→(L))	R(E→(L))	R(E→(L))	
7		R(L→EL)			

(3)

栈	输入	动作
0 -	((a)a(aa))\$	S2
0 2 - (	(a)a(aa))\$	S2
0 2 2 - ((	a)a(aa))\$	S3
0 2 2 3 - ((a	)a(aa))\$	R(E→a)
0 2 2 5 - ((E	)a(aa))\$	R(L→E)
0 2 2 4 - ((L	)a(aa))\$	S6
0 2 2 4 6 - ((L)	a(aa))\$	R(E→(L))
0 2 5 - (E	a(aa))\$	S3
0 2 5 3 - (E a	(aa))\$	R(E→a)
0 2 5 5 - (E E	(aa))\$	S2
0 2 5 5 2 - (E E (	aa))\$	S3

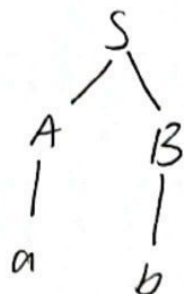
栈	输入	动作
0 2 5 5 2 3 - (E E (a	a)) \$	R(E → a)
0 2 5 5 2 5 - (E E (E	a)) \$	S3
0 2 5 5 2 5 3 - (E E (E a	)) \$	R(E → a)
0 2 5 5 2 5 5 - (E E (E E	)) \$	R(L → E)
0 2 5 5 2 5 7 - (E E (E L	)) \$	R(L → EL)
0 2 5 5 2 4 - (E E (L	)) \$	S6
0 2 5 5 2 4 6 - (E E (L)	) \$	R(E → (L))
0 2 5 5 5 - (E E E	) \$	R(L → E)
0 2 5 5 7 - (E E L	) \$	R(L → EL)
0 2 5 7 - (E L	) \$	R(L → EL)
0 2 4 - (L	) \$	S6
0 2 4 6 - (L)	\$	R(E → (L))
0 2 1 - E	\$	ACC

(4) 首先构造相应的LR(0)项目集规范族

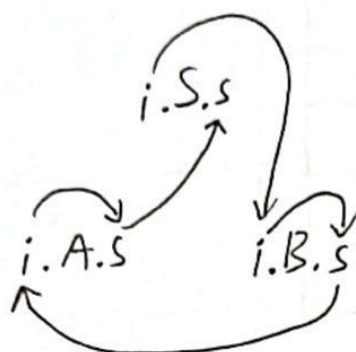


	action				goto	
	(	)	a	\$	E	L
0	S26		S37		1	
1				ACC		
26	S26		S37		5	410
37	R(E→a)	R(E→a)	R(E→a)	R(E→a)		
410		S811				
5	S26	R(L→E)	S37		5	9
811	R(E→(L))	R(E→(L))	R(E→(L))	R(E→(L))		
9		R(L→EL)				

三、(1)



(2)

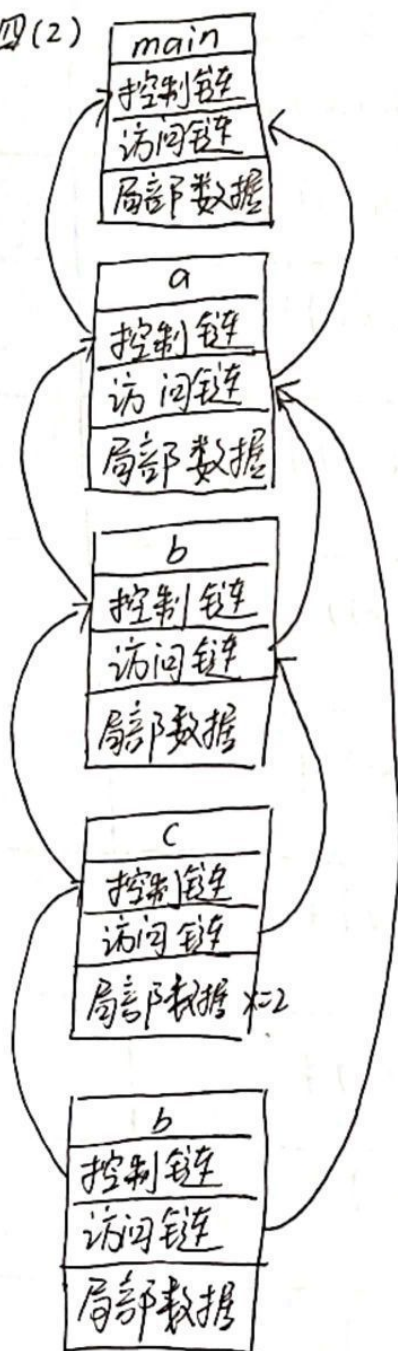


名字x对于过程c来说是非局部名字，过程c沿着访问链前进两步，到达过程a的活动记录。名字x的存储空间就在该活动记录的局部数据区中。然后，根据名字x的存储地址（保存在符号表中）即可访问名字x的存储空间。

(3) $B.i = S.i$
 $B.S = B.i + 4$
 $A.i = B.S$
 $A.S = A.i + 3$
 $S.S = A.S$

四(2)

(3)



(4) 17

四(1)

